

TPMS Explorer: an interactive, gate-verified open ecosystem for triply periodic minimal surface design, generative implicit fields, multifunctional physics simulation, and biophysical tissue-ingrowth dynamics

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Abstract

We present TPMS Explorer v1.0.0, a browser-based, zero-backend ecosystem for parametric design, physics simulation, and additive-manufacturing preparation of triply periodic minimal surface (TPMS) architected materials. The platform couples a watertight Surface Nets meshing pipeline with forty-six continuously executed CI gates ($> 1,000$ assertions under strict per-line accounting, each gate carrying a minimum-assertion guard) that re-execute the exact code claim shipped to the user. The surface library spans 24 level-set families — the eight canonical topologies plus C2 extensions (O,C-TO; Karcher; Fischer-Koch S/Y/C(S)/C(Y); G'; D'; Double P/D/G; Complementary D; Slotted P, F, Q*, W) transcribed from the MiniSurf reference implementation and the jwf23 equation-based lattice dataset, with four-way formula parity (TypeScript / Python / MATLAB / GPU IR) and a public geometric benchmark matrix. Version 7.0 adds five research-grade modules: (i) a sinusoidal-activation implicit neural field (SIREN) with an exactly periodic Fourier-feature input and an 8-dimensional latent space over five distilled TPMS experts; (ii) multiaxial anisotropic yield envelopes (Hill-48, Tsai-Wu, Gurson, Drucker-Prager) with a unified star-convex ray-distance formulation and an interactive 3D viewer; (iii) a Bloch-Floquet phononic band-structure solver built on Born-von Kármán lattice dynamics with realified two-pass deflate-Lanczos eigensolution and long-wave dynamical-matrix calibration; (iv) a reaction-diffusion tissue-ingrowth engine coupling quasi-steady oxygen transport, hypoxia-gated osteoblast proliferation, and mineral deposition over a 28-day horizon; and (v) a level-set topology-optimization loop with Hamilton-Jacobi Godunov evolution driven by elastic and flow sensitivities under a volume constraint. Post-release hardening adds (vi) a deterministic exact-porosity solver — root finding with deterministic Monte-Carlo integration of the analytic surface, with mesh-measured secant validation, reaching 0.13 pp deviation at resolution 96 — and (vii) constructive global orientation propagation that reduces misoriented STL edges to zero by construction, replacing a per-triangle normal-voting scheme that degraded winding consistency five-fold. Version 8.0 adds (viii) an agentic verification layer: a five-operation deterministic tool schema consumed through LLM function calling under a per-slot validating interceptor (enumeration, range, unknown-property, and path-traversal rejection), and a closed-loop design-verification driver in which the LLM selects repairs only from a bounded strategy menu while patch application, execution, and acceptance remain deterministic — including structured unreachableability declarations for representation-limited parameter combinations. The verification suite also audits itself: a meta-gate scans all gate sources for tautological assertions and guard-baseline drift, and the WGSF compute backend is executed, not merely type-checked, on both a physical GPU and a CI software rasterizer (worst-case field deviation 1.9×10^{-4} against a double-precision reference). In live trials the repair menu exhibits both bounded behaviours as intended: given a representation-limited (family, container) combination the model declares structured unreachableability rather than blind-patching, and given an enumeration typo it applies the minimal single-field patch, converging to a watertight STL in one repair round. Versions 9.0–9.1 close the manufacturing loop: (ix) conformal filling of arbitrary anatomical manifolds (voxelized signed-distance containers with wall-side Gaussian deburring and a divergence-volume porosity denominator); (x) ISO 13314 experimental-curve inversion of universal-testing-machine CSV exports with Toe virtual-origin compensation and Gibson-Ashby calibration ratios; (xi) direct implicit slicing — scan-line interval extraction skipping the triangle mesh entirely, emitting layered SVG and the Common Layer Interface industrial format with $\leq 0.1\%$ hatch-volume fidelity — plus an overhang-angle printability audit with deterministic build-orientation search; and (xii) a runnable-OpenFOAM delivery chain (SIMPLE steady dictionaries with embedded pressure and wall-shear probes) consumed by a two-point Forchheimer post-processor yielding an intrinsic permeability of $2.34 \times 10^{-9} \text{ m}^2$ on a measured WSL run. A radial-gradient geometry family (center-dilated metric remap with wall-thickness compensation) motivated a second mesh extractor — Marching Tetrahedra with corner-value regularization and a shape-invariant degeneracy criterion — extending watertight output to non-smooth implicit fields that defeat Surface Nets. All simulation modules execute client-side with zero dependencies beyond Three.js; the agent layer runs under Node.js and exports to industrial formats (STL, 3MF, GLB, Abaqus INP, OpenFOAM polyMesh, G-code), guarded throughout by behavior-level audit gates whose calibration traps are documented.

Keywords: TPMS, implicit neural representation, SIREN, yield surface, phononic bandgap, tissue engineering, level-set topology optimization, LLM tool calling, continuous integration

1. Impact overview

Triply periodic minimal surfaces — first catalogued systematically by Schoen Schoen (1970) — are today a mainstay of architected-materials design and bone-scaffold engineering, where matching a target porosity while keeping the lattice manufacturable is the central numerical problem. TPMS Explorer turns a browser tab into a research bench for architected materials. The five v7.0 modules extend the platform from “geometry plus linear physics surrogates” to generative design, structural-failure envelopes, wave dynamics, biophysical growth, and closed-loop shape optimization — each guarded by an audit gate that executes the same code paths the user sees (Table 1). Version 8.0 completes the loop on the automation side: an agentic verification layer turns natural-language design intents into schema-validated tool calls and drives verification–repair cycles to convergence, with every acceptance decision still taken by deterministic code (Section 3.8). Versions 9.0–9.1 extend the loop to the physical side: anatomical conformal filling, experimental-curve inversion, direct implicit slicing with industrial CLI output, printability audit, a runnable-OpenFOAM permeability chain, and a dual-extractor meshing architecture (Section 3.9).

2. Necessary resources

2.1. Hardware

Any 2020-class laptop. WebGPU is optional (acceleration only); all v7.0 modules run on the CPU path.

2.2. Software

Modern Chromium/Firefox/Safari; Node.js ≥ 20 for the CI suite; TypeScript 6, Vite 8, Three.js 0.185 (only runtime dependency).

3. Method design

3.1. SIREN implicit field with exact periodicity

A 4-layer sinusoidal network in the spirit of SIREN Sitzmann (2020), $F_\theta(\mathbf{x}) = W_3 \sin(W_2 \sin(W_1 \sin(W_0 \mathbf{u}))) + \mathbf{b}_3$ consumes the Fourier feature $\mathbf{u} = [\sin x, \cos x, \sin y, \cos y, \sin z, \cos z]$. Because every coordinate enters only through $\sin/\cos$, F_θ is *exactly* 2π -periodic (measured deviation 2.2×10^{-15}), which makes cross-period water-tightness a property of the architecture rather than of the fit. Five experts (Gyroid, Diamond, Lidenoid, Schwarz P, fractal trabecular) are distilled offline from the platform’s single semantic source of TPMS fields with Adam (RMSE 0.69%–8.04% at 16^3 , hardest for the multi-scale teacher). A latent code $\mathbf{z} \in \mathbb{R}^8$ selects experts through $m = \text{softmax}(-\|\mathbf{z} - \mathbf{A}_i\|^2/2\sigma^2)$ with ± 3 Walsh anchor codes (pairwise Hamming distance 4, cross weight e^{-72}). Lipschitz continuity is bounded by $\prod_i \|W_i\|_2$ (power iteration) and cross-checked against sampled gradient suprema.

3.2. Multiaxial yield envelopes

All four criteria are expressed in a unified *ray-distance* form: the distance $r(\hat{\mathbf{n}})$ from the origin to the envelope along a unit direction of principal-stress space. Hill-48 Hill (1948) (with a hydrostatic cap), capped Drucker–Prager, and Gurson close to star-convex bodies containing the origin, hence r is single-valued and the safety factor is exactly $r(\hat{\mathbf{n}})/\|\boldsymbol{\sigma}_0\|$. Tsai-Wu admits a closed-form quadratic root. Gurson’s hydrostatic pole Gurson (1977) $(2\sigma_0/3) \text{arccosh}((1+q^2)/2q)$ with q equal to the porosity provides a strong analytic anchor. Scaffold-level parameters are derived from the same Gibson–Ashby constants Gibson (1997) used by the platform’s energy-absorption module.

3.3. Bloch–Floquet phononic bands

The unit cell is discretized into voxel masses connected by 26-neighbourhood axial springs; the Bloch–Floquet treatment follows the standard phononic-crystal formalism Hussein (2014). Bloch phases assemble a complex Hermitian $K(\mathbf{k})$, realified as $\begin{pmatrix} A & -B \\ B & A \end{pmatrix}$ so that all eigensolution machinery is real. Two-pass *deflate-Lanczos* (full reorthogonalization; the second run is deflated against the first run’s Ritz vectors) recovers both copies of every complex eigenvalue, preserving degeneracies. Spring stiffness κ is calibrated by exact inversion of the long-wave dynamical matrix $\omega_{\text{LA}}^2 = \kappa h^2 k^2 S_x / (mN^3)$, $S_x = \sum dx^4/d^2$ — affine homogenization is explicitly *not* the long-wave modulus of a central-force network (measured ratio 0.53, matching the analytic 0.54). Under periodic boundary conditions rotations are not admissible Bloch modes, so the Γ point carries exactly three translational zero modes.

3.4. Tissue ingrowth reaction–diffusion

Oxygen obeys a quasi-steady diffusion–Michaelis–Menten balance solved by 7-point Jacobi iteration with no-flux mirrors at solid interfaces and Dirichlet C_0 on the perfused boundary layer; osteoblasts follow hypoxia-gated logistic growth with explicit daily updates; mineral accumulates where oxygen exceeds a threshold. Operator splitting is justified by $\tau_{O_2} = L^2/D \approx 0.05$ days versus a 28-day proliferation horizon. V_{max} is calibrated to a literature oxygen penetration depth $L_{\text{pen}} = \sqrt{2DC_0/V_{\text{max}}} \approx 0.4$ mm. The gate verifies the discrete mass balance (boundary-layer flux = interior consumption, residual 0.000%) and hypoxia gating on a dense-versus-porous pair.

3.5. Level-set topology optimization

The TPMS signed field evolves under $\partial\Phi/\partial t + V_n|\nabla\Phi| = 0$ (Godunov upwinding) with V_n assembled from elastic von-Mises sensitivities (fully-stressed) and Darcy pressure-gradient sensitivities, zero-measured over *interface* voxels, normalized, and constrained by a volume Lagrange bias. Classical compliance under a unit service load is proportional to $1/\text{reaction of the displacement-controlled solve}$ — the $\frac{1}{2}F\delta$ reading is a softening artefact under displacement control. Islands are removed by largest-component BFS, Φ is reinitialized by an exact 3D Euclidean distance transform, and the final field re-enters the platform’s Surface Nets pipeline through the precomputed-field injection contract, guaranteeing water-tight extraction.

3.6. Exact porosity solver: Monte-Carlo integration root finding

The default porosity path couples the requested value to a voxel-rank bisection whose caliber differs from the delivered triangle mesh; measured deviations reached 24 pp for higher-harmonic families at coarse resolutions. Following the integration methodology of RegionTPMS Karakoç (2021), the agent CLI now (i) locates iso^* on the *analytic* field by bisection over a deterministic Monte-Carlo integration of $P(F < iso)$ in the unit cell (LCG-seeded, bit-reproducible), then (ii) builds once with iso^* (bypassing the internal bisection) and applies a single secant correction along the analytic slope to absorb the mesh-versus-analytic volume gap, reverting to the direct build if the corrected attempt is worse. Measured deviations: 0.13 pp (Diamond, 60% target, resolution 96) and 1.3–4.3 pp across cubic-symmetric families at resolution 48; higher-harmonic families at resolution 48 remain representation-limited (9–23 pp) and are reported honestly with an upgrade hint. A cube + Diamond + resolution-64 combination with non-manifold edges (nm = 740, down from 1940 under the legacy path) is fail-closed by the water-tightness gate.

3.7. Global orientation propagation

The binary and multi-solid STL writers previously flipped each triangle independently by comparing geometric normals against the vertex normal field; because that field is unreliable at cap fans and boundary skins, independent voting fragmented a locally consistent winding (misoriented edges 4536 → 23571 after flipping at resolution 96). The writers now call a constructive orientation pass: register every undirected edge with its two incident triangles, two-color the triangle adjacency graph by BFS (same-direction neighbours get opposite colors), apply the parity, and flip the whole mesh once if the divergence-theorem volume is negative. For the closed manifolds the pipeline guarantees (open = non-manifold = degenerate = 0), misoriented edges are zero by construction — verified byte-level on the exported STL (823,500 edges at resolution 96). Figure 1 contrasts the two writer strategies on identical geometry (F-RD lattice, resolution 96, $k=6$, 838,008 triangles): the extraction output itself carries 15,552 misoriented edges concentrated along high-curvature sheet junctions; the historical per-triangle voting pass not only fails to repair them but fragments the winding across the entire surface (16,104); the constructive propagation pass eliminates all of them. The same experiment on the earlier extraction pipeline recorded 4,536 → 23,571 under voting — the scheme degrades winding whenever the vertex-normal field it votes against is unreliable.

3.8. Agentic closed-loop verification (v8.0)

The agent layer follows one iron rule: the LLM only fills schema-bounded slots, and every numeric value that reaches the geometry pipeline is validated (reject-on-violation) by a deterministic interceptor — enumeration membership, numeric ranges, unknown-property rejection, and single-segment path patterns (anti path-traversal) are enforced per slot, so adversarial or malformed model output can neither inject code nor escape the parameter domain regardless of model behavior. Five

operations are registered in the tool schema: *list*, *estimate*, *mesh* (direct watertight build), *scenario* (end-to-end scaffold delivery: STL + Abaqus INP + verification report), and *design-verify* (the closed-loop entry). A natural-language agent maps bilingual instructions — with alias resolution such as Schwarz D → diamond — to tool calls through LLM function calling over three interchangeable providers: a local Ollama endpoint, any OpenAI-compatible endpoint, and an offline Mock provider whose real-execution path is guard-railed behind an explicit opt-in so that closed-loop regression can run without network access.

The design-verify driver closes the loop: *propose* → *verify* (water-tightness, porosity-deviation, and physical-range checks) → on failure, the LLM selects the next repair *only* from a bounded strategy menu (surface-family swap, period reduction, resolution escalation ≤ 128 , container or mode change, per-slot parameter fixes) → deterministic patch application and re-run. Repairs inside the existing verification ladder (automatic resolution escalation, secant correction) remain deterministic; the driver handles ladder-external repairs, and for representation-limited combinations (e.g. the cylinder + diamond deep-water region identified by the benchmark matrix) it emits a structured *unreachability declaration* instead of looping. Acceptance rests on three layers: a 37-instruction bilingual regression against a production LLM (glm-5.3-flash multi-round 37/37 (C3 mitigated, $n \geq 2$; not deterministic); all adversarial samples rejected or self-clamped; a four-tier model profile documents server-side variance as the residual failure mode, never the pipeline); an offline mock-driven closed-loop self-test covering parameter-fix convergence, out-of-ladder resolution repair, period reduction, unreachable declaration, interceptor rejection, and round exhaustion with the original design file bit-identical; and a 106-assertion schema↔CLI cross-check gate that pins enumeration parity across tools, a static sentinel between the schema type enumeration and the driver’s repair menu, and an offline end-to-end execution of the bridge. The LLM regression itself is key-gated and runs outside CI; correctness of the layer does not depend on it because the interceptor is deterministic. Figure 2 summarizes the resulting closed-loop state machine.

3.9. The v9.x manufacturing loop: from anatomy to permeability

The v9.x releases extend the verified chain from geometry to physical deliverables, each link carrying its own analytic anchor. *Conformal filling* (v9.0) voxelizes any watertight anatomical STL into a signed-distance container (welded vertices, weighted ray crossing, uniform-grid nearest-surface buckets), fuses it with the TPMS field through a wall-side Gaussian deburr that preserves the hard air override outside the container, and defines porosity only over the container’s divergence-theorem volume — eliminating a +30 pp envelope artefact measured for AABB-box denominators. *Experimental-curve inversion* maps universal-testing-machine CSV exports to ISO 13314 features (toe virtual-origin compensation, union-of-bands slope substitution, Rp0.2, plateau stress) with Gibson–Ashby calibration ratios, validated against independently computed reference values. *Direct implicit slicing* extracts solid intervals per scan line directly from the implicit field — no triangle mesh

Misoriented edge midpoints, F-RD lattice (R96, $k=6$, 838,008 triangles)

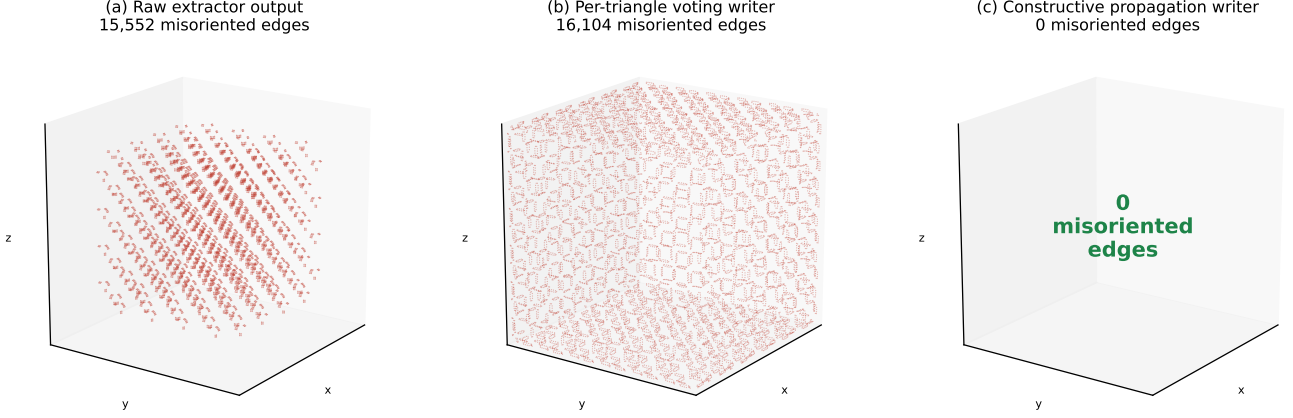


Figure 1: Orientation states of the same F-RD lattice (R96, $k=6$) under three writer strategies, with misoriented-edge midpoints overlaid on the unit cell. (a) Raw extraction output: 15,552 misoriented edges cluster along sheet junctions. (b) The historical per-triangle voting writer (flip each triangle whose winding disagrees with the summed vertex normal) raises the count to 16,104 and scatters the defects over the whole surface. (c) The constructive propagation writer (undirected edge registration, BFS two-coloring of the triangle adjacency graph, and a divergence-volume sign flip) yields zero misoriented edges by construction.

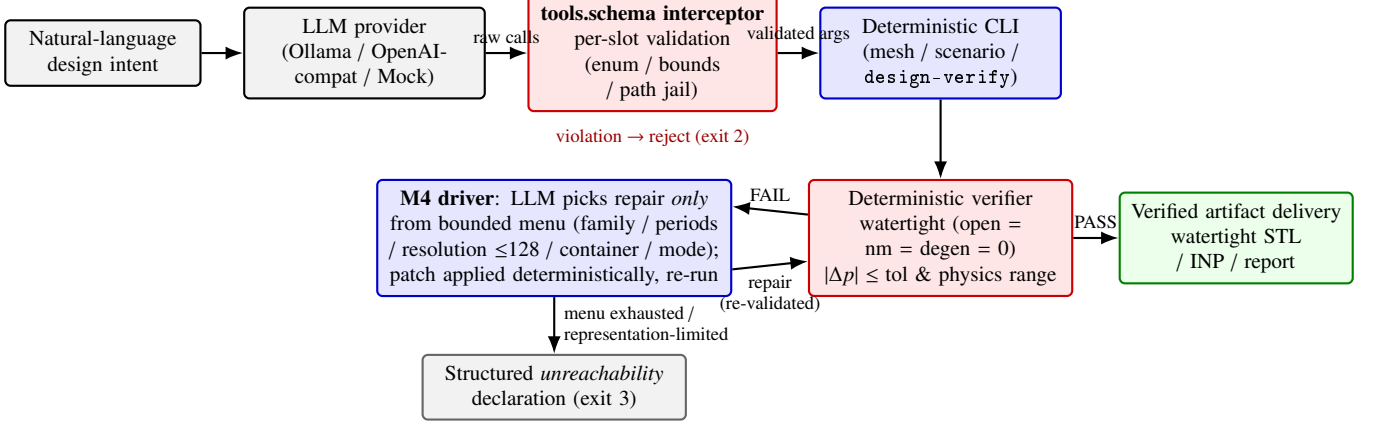


Figure 2: The agentic closed loop (v8.0). The LLM is confined to proposing schema slots: a deterministic interceptor validates every call (reject on violation) before execution, the verification ladder is purely deterministic, and repair decisions are re-validated from a bounded strategy menu — yielding either a verified watertight artifact or a structured unreachable declaration, never a silently degraded mesh.

— with volume parity $\leq 2\%$ against the mesh divergence measure and $\leq 0.1\%$ hatch-volume fidelity after Common-Layer-Interface serialization. The accompanying *printability audit* reports overhang statistics under the industrial convention $\alpha = \arccos(-\mathbf{N} \cdot \mathbf{b})$ with a deterministic Fibonacci-sphere orientation search; six measured specimens show TPMS lattices to be near isotropic (orientation gain $\approx \pm 1$ pp), quantifying when slicer-side support control suffices. The *CFD delivery chain* emits a runnable OpenFOAM case (SIMPLE steady, flow-rate inlet, kinematic pressure, embedded area-averaged pressure and wall-shear probes, Robin oxygen walls under a self-consistent mm unit system); a two-point Forchheimer post-processor separates $\Delta P = AQ + BQ^2$ and reports the intrinsic permeability as the Stokes intercept — measured $2.34 \times 10^{-9} \text{ m}^2$ on a WSL run, inside the bone-scaffold literature band, with the inertial fraction rising from 2.6% to 21.1% between the two flow rates. Finally, a *radial-gradient family* (center-dilated metric remap,

$K \in [1, 3]$, arctanh radial and rational axial channels with wall-thickness compensation) exposes fields that are non-smooth at $|P|$ creases and max ridges, where Surface Nets' vertex interpolation is structurally non-manifold; a Marching Tetrahedra extractor with shared-corner ring ordering, corner-value regularization ($\eta = 3 \times 10^{-3}$), and a shape-invariant degeneracy criterion (area/maxedge²) restores watertight output for all $K = 1-2$ (sphere anchor $4\pi/3$ within 0.13%); a one-shot UI card mirrors the CLI pipeline (single-shot preview at $R=48$, HD STL export at $R=96$) behind a rebuild-invalidation guard that prevents stale asynchronous geometry from overwriting the preview, giving the platform a *dual-extractor* architecture: Surface Nets for smooth fields, MT for the rest.

4. Quality control

Forty-four gates run via The C2 extension families are transcribed from the MiniSurf reference implementation Hsieh (2020)

and the equation-based lattice dataset of Fleck et al. Fleck (2023) (CC BY 4.0). `npm run test:all` on three platforms. Beyond CI, the exported MATLAB reconstruction scripts are executed on a real MATLAB R2025a runtime as a standalone three-branch smoke (default identity-manifold, second-harmonic, and non-Euclidean Poincaré geometries) — the probe caught a latent `isonormals` failure unique to deformed (non-separable) grids, since fixed by gating the normal interpolation on separable meshes. Every gate loads the shipped TypeScript through a bundler and asserts behavior-level invariants (no mocked math), and every gate now carries a *minimum-assertion guard*: a probe experiment showed that an auditor whose assertions are all neutralized still exits green under the historical `if (fail)` exit idiom, so each gate additionally fails when its executed-assertion count drops below a measured baseline. Three agent-layer self-checks whose unregistered periods had incident histories of silent redness (a drifted fail-closed contract that went unnoticed for a day) were promoted into the CI scheduler so that the same discipline covers the automation layer itself. Four tautological assertions (two `|| true` disjunctions, a self-multiplying metric check, and a dead variable that was computed but never asserted) were exposed by a dedicated audit and either removed or rewritten as real checks; one of them had been masking a wrong closed-form slope in the Poincaré extension for its entire lifetime. Gate calibration is treated as a first-class activity: documented false-red episodes (a $\sqrt{2}$ scaling slip in the auditor von-Mises conversion; a Gram–Schmidt normalization threshold that silently skipped 10^{-10} -scale columns; a cross-implementation static sentinel left unsynced after a formula change) are recorded and their fixes covered by regression assertions.

5. Availability

5.1. Current code version

v1.0.0-stable-epoch (MIT license; version history: prototype-era releases through v9.2, semantic stable series from v1.0.0).

5.2. Operating requirements

See Necessary resources; no backend, no telemetry, no build-time secrets.

5.3. Installation

Open `docs/platform/index.html` (static build) or run `npm install` and `npm run dev` in `tpms/tpms-platform`. CI: `npm run test:all`. A deterministic agent CLI (`tpms/agent/tpms.mjs`) exposes `list`, `estimate`, `mesh`, `verify`, `solve`, `scenario`, `slice` (SVG/CLI layered output), `overhang` (printability audit), and `cfd-post` (Forchheimer permeability) subcommands with JSON output for scripting; `llm-agent.mjs` adds natural-language tool calling and `tpms-driver.mjs` closed-loop design verification (LLM API keys are supplied via environment variables and never committed).

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Table 1: v7.0–v9.2 modules and their CI gates.

Stage	Module	Verified invariants	Assertions
I	SIREN implicit field	exact 2π periodicity (2.2×10^{-15}); distillation RMSE per expert; Lipschitz product bound vs. sampled gradients; latent-path Lipschitz; water-tight meshes at anchors and manifold midpoints	43
II	Yield envelopes	uniaxial analytic anchors; midpoint convexity ($\times 200$ per criterion); isotropic permutation symmetry; Gurson hydrostatic pole; closed-mesh volume by divergence theorem; safety factor of a half-scaled stress = 2	35
III	Phononic bands	Γ -point triple translational zero modes; sound-speed self-consistency (gyroid 2.2%); per- k branch ordering; time-reversal symmetry $\omega(\mathbf{k}) = \omega(-\mathbf{k})$; diatomic-chain analytic gap; island-count guard	18
IV	Tissue ingrowth	mass balance = boundary-layer flux – interior consumption (0.000%); core-oxygen transport differentiation; $\rho \leq \rho_{\max}$; hypoxia gating; mineral monotonicity; bitwise determinism	11
V	Level-set optimization	unit-load compliance $\propto 1/\text{reaction}$ monotone under 0.5% tolerance; volume Lagrange constraint $\pm 10\%$; permeability gain $\times 1.50$; water-tight extraction (open edges = 0); reinitialization $ \nabla\Phi \approx 1$	8
VI	Agentic verification (v8.0)	schema $\leftrightarrow$ CLI cross-check (enum/range/path parity, incl. static sentinel vs. driver repair menu); interceptor validation incl. adversarial and malformed calls; mock-driven closed-loop convergence on injected defects with the original design file bit-identical	126
VII	Manufacturing loop (v9.0–9.1)	conformal-fill SDF vs. analytic anchors and divergence-volume denominator; ISO 13314 inversion against independent reference values; slicing volume parity vs. mesh divergence ($\leq 2\%$) and CLI hatch fidelity ($\leq 0.1\%$); overhang spherical-cap integral anchor ($14.645\% \pm 1\%$); Forchheimer synthetic exact-recovery anchors; MT sphere anchor ($4\pi/3$ within 0.13% at R48)	66+32+18